MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Aeronautical/Automobile Engineering**	**Course Type**	**FC**
Course Name	**Industrial IoT**	**Course Code**	**AAE - 3124**
Semester	**V**	**Max Marks**	**50**
Date	**30.11.2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	How do Cyber Manufacturing Systems (CMS) seamlessly integrate physical and digital components in manufacturing? Discuss the working procedure with an example.	4	1	1,2	1	1,2,4,5	L2
1B	Analyze how industries can leverage Big Data Analytics to enhance engine failure prediction. Explain the process in detail and support your discussion with relevant examples.	4	1	1,2	1	1,2,4,5	L4
1C	Interpret the concept of Service-Oriented Architecture (SOA) and explain how it enables communication between diverse services within a network.	2	1	1,2	1	1,2,4,5	L3
2A	How do the various components of the multi-mover system interact with the motor to ensure effective operation? Provide a detailed explanation of these interactions with a neat sketch.	4	2	1,2	2	2,3,5,6	L2
2B	Interpret the concept of a hybrid automaton and explain the role of the Localization Subsystem (LOC) in ensuring the overall functionality of the multi-	3	2	1,2	2	2,3,5,6	L3

	mover system. Support your explanation with a relevant diagram.						
2C	Apply the hybrid automaton model to analyze the Battery Subsystem (BA) within the multi-mover architecture. Discuss its functionality and role in the system.	**3**	**2**	**1,2**	**2**	**2,3,5,6**	**L3**
3A	What are the main components of a Cyber-Physical Production System (CPPS), and how do they work together to create an integrated production system? Explain with the help of a relevant diagram.	**4**	**3**	**1,2,3**	**3**	**2,3,5,6**	**L2**
3B	Interpret the role of wireless communication in enhancing the functionality of the Industrial Internet of Things (IIoT). Discuss the various wireless communication technologies and analyze how they address critical factors such as range, power consumption, bandwidth, latency, and reliability in industrial environments.	**4**	**3**	**1,2,3**	**3**	**2,3,5,6**	**L3**
3C	Demonstrate the operational mechanism of star topology in an Industrial Internet of Things (IIoT) environment, highlighting its functionality and advantages.	**2**	**3**	**1,2,3**	**3**	**2,3,5,6**	**L3**
4A	Explain the classification model for the manufacturing environment with the help of a relevant diagram. Support your explanation with practical example.	**4**	**4**	**1,2,3,5**	**4**	**2,3,5,6,12**	**L2**
4B	Describe the working flow of big data architecture with a relevant block diagram. Interpret this flow using smart agriculture as an example.	**4**	**4**	**1,2,3,5**	**4**	**2,3,5,6,12**	**L3**
4C	Discuss the principles of hard and soft real-time requirements within the framework of Industrial Internet of Things (IIoT) applications.	**2**	**4**	**1,2,3,5**	**4**	**2,3,5,6,12**	**L2**
5A	Discuss the efficiency of an IoT-based smart metering system in comparison to conventional systems.	**4**	**5**	**1,2,5,12**	**5**	**1,2,3,18**	**L4**

	Examine the internal architecture of a smart metering system with the help of a relevant diagram.						
5B	Analyze how IoT transforms conventional homes into smart homes, emphasizing the architectural layers of smart automation. Support your analysis with a detailed and well-illustrated diagram.	**4**	**5**	**1,2,5,12**	**5**	**1,2,3,18**	**L4**
5C	Apply the methodology of smart card assembly to outline the step-by-step process for creating a contact-based smart card, with a focus on integrating its essential components.	**2**	**5**	**1,2,5,12**	**5**	**1,2,3,18**	**L3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**10**
2	**10**
3	**10**
4	**10**
5	**10**

BL	Total Marks
2	**18**
3	**20**
4	**12**
5	